

Predicting the NFL Draft

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Abstract

The NFL Draft is a pivotal event where teams select eligible college football players, yet traditional scouting methods rely heavily on subjective analysis and speculation. This study introduces a data-driven approach to draft prediction, leveraging machine learning techniques to analyze historical draft outcomes, NFL Combine metrics, and college performance statistics. We employ XGBoost, neural networks, and random forests to predict both first-round draft likelihood and the specific round in which a player will be selected. Our results demonstrate the effectiveness of machine learning in enhancing draft predictions, providing an objective alternative to traditional scouting methodologies.

Introduction

For many football fans, the months following the NFL season can feel slow, with little excitement until the NFL draft arrives. The NFL Draft is an annual event where NFL teams select eligible college football players to join a team roster. The draft is composed of seven rounds, and each team gets one pick per round, though trades can alter the order. Players are evaluated by NFL coaches at the NFL Combine event, a week-long event that takes place in Indianapolis, Indiana. They're evaluated on their strength and athleticism in key events like 40-yard dash, bench press, vertical leap, shuttle drills, three-cone drill, etc. The information gathered during the combine plays a critical role in how NFL teams evaluate players and make their selections during the draft. [1]

Without a strong background in college football, it's difficult to engage in draft discussions or predict player success at the next level. Traditional mock drafts are often based on speculation and subjective opinions rather than rigorous data analysis. Sports analysts and media personalities frequently rely on limited game footage or surface-level evaluations, leading to inaccurate projections.[2] This project aims to take a different approach—leveraging machine learning and historical data to build a data-driven model for predicting draft outcomes.

By systematically analyzing past drafts, NFL Combine results, and college statistics, we make it our objective to create a model that provides an objective assessment of a player's likelihood of being drafted in the first round. Unlike traditional scouting reports, which can be influenced by biases and narratives, this approach focuses purely on measurable performance metrics.

Dataset & Features

Our dataset is comprised of 13381 players, with an 80-10-10 split between training, test, and validation sets. Our features include college statistics and combine measurements. Of the available statistics, we only took into account count statistics, so all ratings and averages were not considered. This means we also face issues with missing college stats, as around 2000 of our 13,000 total players did not have their stats available. To the model, these players look like they did not accumulate any stats in college. Table 1 has some example information we have on Quarterback Drew Brees when he was a draft prospect.

For combine measurements, we took into account physical metrics such as a player's height or 40-yard dash time. Unfortunately, not all college players are invited to the combine, and even those that are opt out of some if not all exercises. To fill these missing measurements, we used `IterativeImputer` from `sklearn` to fill in missing data. This imputes values based on relationships between existing measurements.

Table 1: Example Draft Data for Drew Brees (Certain stats and measurements omitted for space)

Attribute	Value	Attribute	Value
Player	Drew Brees	Height (inches) [†]	73.08
Pick	32.0	Weight (lbs) [†]	234.34
Position	QB	40-yard dash (sec) [†]	4.77
College	Purdue	Vertical jump (inches) [†]	32.14
Draft Year	2001.0	Bench press (reps) [†]	19
Age	22.0	College games [◇]	41.0
Seasons [◇]	4.0	Pass attempts [◇]	1525.0
Pass completions [◇]	942.0	Pass yards [◇]	10909.0
Pass TDs [◇]	81.0	Interceptions [◇]	41.0

[†] Data from NFL Combine measurements

[◇] Data from college statistics

Methods

Data Collection & Processing

To gather data, we used Pro Football Reference [3] and College Football Reference (CFR) [4]. PFR has historical NFL combine and draft data dating back to 2000. CFR has the statistics of almost all NFL prospects. In order to retrieve the raw data from these sites, we needed to scrape the HTML of over 11,000 pages (mostly getting individual player stats).

This was completed using the `requests` Python library with random delays to avoid rate-limits and avoid bot-detection. We also used `Selenium` [5] for JavaScript-rendered content. Scraping all the data took 24 hours in total. In order to not repeat this process in the future, we cached the entire target page's HTML locally so we could re-reference the raw data efficiently.

After caching all of the HTML, we extracted target features using `BeautifulSoup` and saved the result as `csv` and `feather` files using `pandas`. At this point, a player's combine measurements, draft position, and college stats were in 3 different tables, so we created a unique identifier for each player and combined the tables.

Position information was encoded using one-hot encoding across 24 possible NFL positions (e.g., QB, RB, WR, DE). For colleges, we calculated how many players have been historically drafted from a school and used that metric to delineate small, medium, and large schools. We then created an one-hot encoding for those 3 distinctions. Numeric features were standardized using `sklearn's StandardScaler`.

XGBoost

XGBoost is particularly suitable to our draft predictions given its sequential tree-building approach given the non-linear relationships between player metrics and draft outcomes. [6]

To predict whether a player would be drafted in the first round, we used the XGBoost Python package. We leverage `scikit-learn` [7] `GridSearchCV` with three-fold cross-validation to optimize our hyperparameters, namely, tree depth, number of boosting rounds, learning rate, subsampling rates, and column sampling.

Our evaluation metric was the area under the receiver operating characteristic curve (AUC-ROC). The most optimal combination included a maximum tree depth of 3, 100 boosting rounds, a learning rate of 0.1, column sampling set to 1.0, and subsampling set to 0.8.

We also used XGBoost to predict what specific round a player would be drafted in (e.g. Round

1, ..., Round 7, Undrafted). An initial grid-search found the best hyperparameters to be a maximum tree depth of 7, 300 boosting rounds, a learning rate of 0.01, column sampling set to 0.8, and subsampling set to 0.7. A subsequent fine-grained grid search further optimized those hyperparameters to be: tree depth of 6, 350 boosting rounds, 0.02 learning rate, column sampling set to 0.85, and subsampling kept to 0.7. [8]

Neural Network

Neural networks also handle complex non-linear relationships, so we constructed networks to predict both whether a player would be drafted at all and which round of the draft they would land in.

For the simpler binary classification (drafted vs. undrafted), we employed a neural network with a sigmoid output layer, allowing the model to produce probabilities between 0 and 1 indicating how likely it is that a given player is drafted. We combined this final layer with a binary cross-entropy loss function (Eq. 1) to guide the training process.

$$\mathcal{L}_{\text{binary}} = -\frac{1}{N} \sum_{i=1}^N \left(y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right) \quad (1)$$

In the multi-class variant (predicting round), network architecture involves ReLU activations in the hidden layers for non-linearity and a softmax activation in the output layer, which yields a probability distribution over each of the eight draft segments (7 drafting rounds and one bin for undrafted). We trained this model using categorical cross-entropy (Eq. 2) as the loss function.

$$\mathcal{L}_{\text{categorical}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log(\hat{y}_{ic}) \quad (2)$$

In both cases, we use `tensorflow's keras` package to define and train the model.

Random Forest

Random forest is an alternative decision tree algorithm that is also suited to our predictions given its ensemble approach of multiple decision trees. Ergo, we used a random forest model to predict which round of the draft a player would be picked in. [9]

The hyperparameters that can be tuned in this model include the maximum tree depth, the number of weak learners, the minimum number of samples required for a node to be considered for splitting, the minimum number of samples in a leaf, and the maximum number of features considered at each split.

To optimize these parameters, we performed randomized search over a set of hyperparameters, and evaluated each subset using 3-fold cross-validation. Our implementation utilizes sklearn's RandomForestClassifier and RandomizedSearchCV packages. The optimized hyperparameters were: 800 estimators, a max depth of 30, a criterion of entropy, max features being $\sqrt{m}$, a minimum samples split of 10, and a minimum samples in a leaf of 4.

Principal Component Analysis (PCA)

While our prediction models were implemented using the full feature set, we used PCA to explore feature importance. [10] In particular, we were interested in the position-specific importance of certain metrics. For example, are passing yards more important than height for a QB's draft position?

After separating our data points by position, we used the PCA package from sklearn to implement PCA. We determine the primary component which captures the largest portion of variability in our dataset. Inside primary component, we find the features with the highest absolute loadings. These features are deemed "important" as they align most strongly with the primary direction of variation.

Results & Discussion

Accuracy	Precision	Recall	F1 Score
0.955	0.571	0.067	0.119

Table 2: XGBoost First Round Metrics

Predicting the 1st Round

As a preliminary step, we focused on predicting whether a player would be drafted in the first round to validate that our dataset is conducive to machine learning-based NFL draft predictions. Using XGBoost, we were able to achieve 95% accuracy (Table 2). That accuracy is slightly misleading considering the proportion of players picked in the 1st round versus not.

Table 3 offers a different perspective. By training on data pre-2023, we can predict for the entire 2023 Draft Class the probability (fr.hat) that a given player was picked in the first round. Taking the highest 32 probabilities effectively gives our first round predictions for 2023.

Comparing to the actual first round picks that year, our model correctly predicted 40.6% of the selections. While this initial result was promising, several inherent limitations in this approach prompted us to

Table 3: 2023 Draft 1st Round Predictions (correct in bold)

Rank	Player	fr.hat	Pick
1	C.J. Stroud	0.881683	2.0
2	Bryce Young	0.769109	1.0
3	Will Anderson	0.550869	3.0
4	Myles Murphy	0.503579	28.0
5	Brian Branch	0.474406	45.0
6	Marvin Mims	0.469272	63.0
7	Nick Herbig	0.452050	132.0
8	Tuli Tuipulotu	0.452050	54.0
9	Jahmyr Gibbs	0.430007	12.0
10	BJ Ojulari	0.429517	41.0
11	Felix Anudike-Uzomah	0.419108	31.0
12	Kelee Ringo	0.411211	105.0
13	Deuce Vaughn	0.410028	212.0
14	Jalin Hyatt	0.390932	73.0
15	Bryan Bresee	0.384207	29.0
16	Jaylon Jones	0.359845	221.0
17	Christian Gonzalez	0.330977	17.0
18	Kayshon Boutte	0.330394	187.0
19	Clark Phillips	0.329543	113.0
20	Bijan Robinson	0.320024	8.0
21	Jordan Addison	0.297313	23.0
22	Noah Sewell	0.295462	148.0
23	Antonio Johnson	0.278784	160.0
24	Parker Washington	0.276040	185.0
25	Lukas Van Ness	0.269404	13.0
26	Gervon Dexter	0.260778	53.0
27	Calijah Kancey	0.257831	19.0
28	Anton Harrison	0.255717	27.0
29	Adetomiwa Adebawore	0.254400	110.0
30	De'Von Achane	0.249512	84.0
31	Isaiah Foskey	0.246663	40.0
32	Trenton Simpson	0.245401	86.0

Table 4: PCA Results by Position

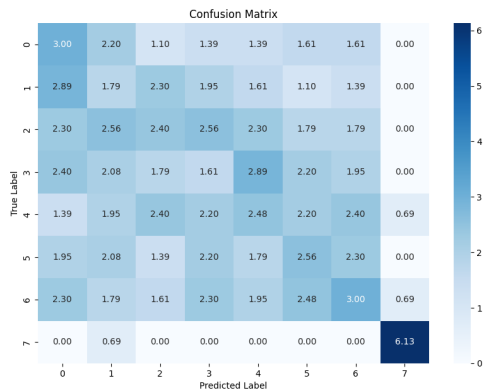
Position	Var. PC1	Top Feature
TE	0.214	Height (0.357)
QB	0.228	Passing Yards (0.273)
DE	0.265	Combined Tackles (0.347)
LS	0.333	Height (0.362)
OG	0.208	40-Yard Dash (0.365)
OT	0.187	Seasons Played (0.333)
T	0.248	Combined Tackles (0.374)
LB	0.221	Broad Jump (0.295)
RB	0.195	Height (0.369)
DT	0.262	Combined Tackles (0.340)
DB	0.216	40-Yard Dash (0.337)
OL	0.228	40-Yard Dash (0.306)
EDGE	0.256	Combined Tackles (0.354)
ILB	0.243	Combined Tackles (0.330)
OLB	0.188	Combined Tackles (0.348)
NT	0.425	Combined Tackles (0.302)
DL	0.223	Tackles for Loss (0.328)

explore alternative draft prediction methods. These limitations include our failure to account for team-specific positional needs and the dependency of draft positions on the relative quality of position peers within the same draft class.

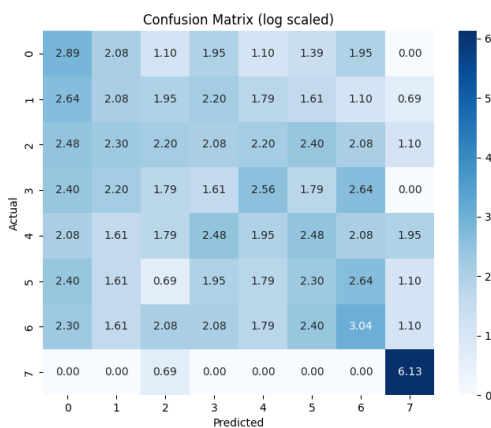
Predicting All Rounds

To extend our analysis beyond the first round, we refined our approach to predict the proportion of players drafted in each of the 7 rounds as well as those that would be left undrafted.

We tackled this as a multi-class classification problem (with label 0 for the first round, label 1 for the second round, ..., and so on, with label 7 representing undrafted players). To model this, we employed three different approaches: a dense neural network, Random Forest, and XGBoost. For all models, we evaluated model performance using four primary metrics: accuracy, precision, recall, and F1 score.



(a) Confusion Matrix for Random Forest



(b) Confusion Matrix for XGBoost

Figure 1: Confusion matrices for Random Forest and XGBoost models.

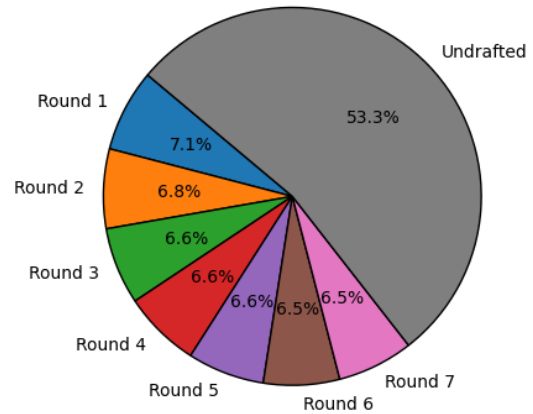


Figure 2: Training Data Label Distribution

Figure 1 shows the confusion matrices for Random forest and XGBoost, which had been log-scaled to better visualize the performance of the models. We can see that the model performs the best on classifying players that did not end up getting drafted, which is encouraging as it reflects the models' ability to distinguish between undrafted players and those selected. However, it's clear that limitations remain in accurately predicting specific draft rounds, particularly in the mid-rounds. Labels 0 (first round) and 6 (last round) show the next highest accuracy after the undrafted category, suggesting some skill in identifying both top-tier prospects and those likely to be drafted towards the end. The discrepancy in predictive power among the drafted and undrafted is likely due to an abundance of data points which are labeled undrafted (Figure 2). The quantitative results for all 3 models can be found in Table 5. All three models showcase a similar accuracy score in the range of 63 – 64%, demonstrating that each is sufficient to explain the underlying patterns within the draft data. Random forest emerged as the best-performing model with an accuracy score of 64.29%, while the neural network model ended up at 63.1%. Despite the relatively improved performance due to the robustness of the decision tree models, all models faced challenges in accurately predicting specific draft rounds, particularly in the mid-range rounds, as reflected by their relatively low precision and recall scores and as displayed in the confusion matrices.

It's clear that compared to predicting for the first round which often features highly distinguishable players with standout statistics and scouting profiles, the mid-range rounds present a significantly greater

Model	Accuracy	Precision	Recall	F1 Score
Neural Network	0.6310	0.2725	0.3000	0.2622
XGBoost	0.6317	0.2841	0.2928	0.2832
Random Forest	0.6429	0.3057	0.3137	0.3058

Table 5: Performance Metrics for Multi Bin Models

challenge. Players drafted in these rounds often have more similar profiles, making it harder for the models to differentiate between them, leading to higher misclassification rates. However, the models perform more strongly towards the final rounds (rounds 6–7) and for players who go undrafted. This is likely because players in these categories exhibit more distinct characteristics that set them apart from earlier-round picks. For example, late-round draftees may have noticeable gaps in their skill sets or athletic profiles, while undrafted players often lack the standout qualities required to be selected at all.

Predicting Drafted vs. Undrafted

As part of the preliminary efforts, a small dense neural network was used to simply predict whether or not a player would be drafted. This model incorporates all features by simply encoding non-numerical features like position and college, to a number. The network involves two feed-forward layers of sizes 16 and 8, both with ReLU activations. We were able to achieve an accuracy of 0.9095. To refine this model, we decided to embed the non-numerical features with an embedding layer. Position is mapped to a 4-dimensional vector while college is embedded to a 8 dimensional vector. This resulted in worse performance, with an accuracy of 0.7821 (see table 6). This could be explained by the fact that we might have insufficient data to learn meaningful embeddings. This is especially true for college, where there are hundreds of unique values but small amount of data per value.

Model	Accuracy
Neural Network w/o Embedding	0.9095
Neural Network w/ Embedding	0.7821

Table 6: Performance Metrics for Binary Models

Position-Specific Feature Importance

Table 4 shows the results of our Principal Component Analysis by position, highlighting the variance explained by the first principal component (PC1) and the "Top Feature" which aligns best with PC1. Notably, the variance explained by PC1 ranges from

18.7% (OT) to 42.5% (NT), indicating considerable differences in how effectively a single dimension captures the draft evaluation criteria across different positions.

The low variance explained by PC1 for certain positions align with the reality that certain evaluations rely heavily on qualitative judgements or position-specific traits. For example, offensive linemen are frequently assessed on technical proficiency (e.g., hand placement, footwork, and blocking technique), as well as position-specific measurements such as arm length which we did not collect. As such, we see very low variance explained by PC1 for the OG (20.8%) and OT (18.7%) positions.

Overall, the relatively low variance explained by PC1 across positions suggests that NFL draft evaluation is inherently multidimensional. Some of those dimensions (technical skills, game intelligence, positional nuances, etc.) are incredibly hard to quantify or even track.

Conclusion

Our results indicate that for multi-bin classification, all models—XGBoost, Random Forest, and Neural Networks—perform similarly, suggesting that predictive accuracy under this context is limited by data quantity and quality rather than model choice. Deep learning, including embeddings, did not provide a clear advantage even in simple binary classification, reinforcing the effectiveness of traditional feature engineering given a somewhat small dataset.

To improve predictions, future work will explore time-series modeling with LSTMs to capture season-by-season performance. Additionally, incorporating team-specific draft tendencies and other tracking metrics like digital foot prints may enhance predictive power. Ultimately, better data, rather than more complex models, is key to refining NFL Draft predictions.

Contributions

- Alastair Deng: Scraped the dataset. Performed all data preprocessing. Implemented XGBoost for predicting the 1st Round. Implemented PCA for determining position-specific feature importance. Edited Introduction. Wrote Dataset & Fea-

tures. Wrote Methods. Wrote the Predicting the 1st Round subsection of Results & Discussion. Wrote the Position-Specific Feature Importance subsection of Results & Discussion.

- Dev Jayram: Implemented Random Forest model for multi bin classification. Implemented a randomized search for Random forest hyperparameter optimization and evaluated each subset of parameters using 3-fold cross validation. Conducted research into LSTM time-series model to improve predictions as a step towards future work as well as an attempted implementation. Wrote Predicting All Rounds subsection of Results & Discussion, Introduction, and Abstract. Edited the Random Forest subsection of Methods.
- Michael Liu: Implemented XGBoost for multi bin classification. Implemented Grid Search for XGBoost hyperparameter optimization. Implemented neural network with and without embedding layer for both drafted vs undrafted and multi bin classification. Wrote Predicting Drafted vs. Undrafted subsection of Results & Discussion. Wrote conclusion. Edited Predicting All Rounds subsection of Results & Discussion.

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